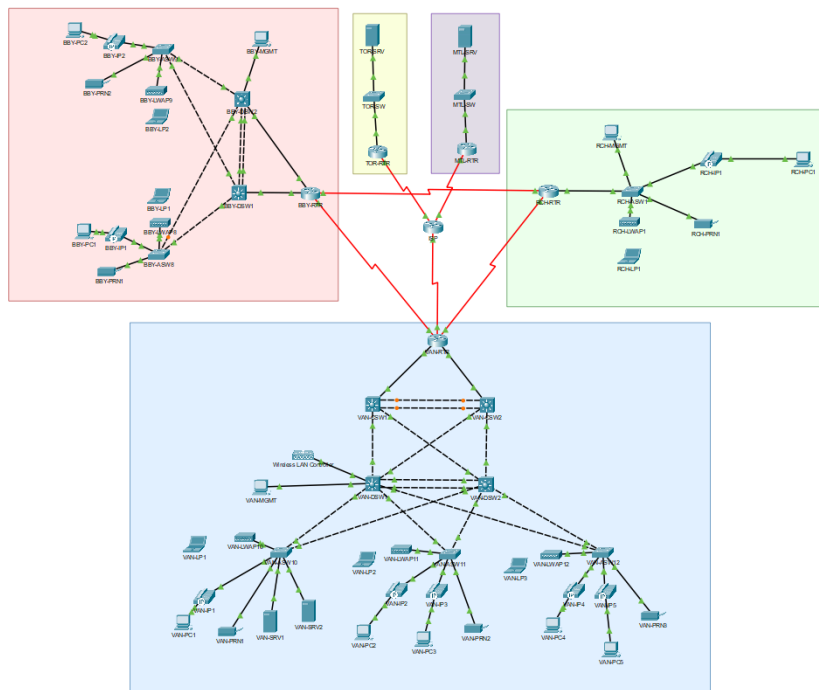


CCNA SPEEDRUN: LAB DOCUMENTATION

(Packet Tracer)



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Phase 1: Basic Device Initialisation

Configure in **ALL NETWORK DEVICES** except Toronto and Montreal devices. You don't need to worry about Toronto and Montreal.

- 1) Configure GLOBAL ENCRYPTION of plaintext passwords
- 2) Configure the hostname of each network device
- 3) Configure **"speedrun.ccna"** as the domain name
- 4) Disable DNS domain lookups
- 5) Configure a MOTD legal banner to give a legal warning for unauthorised users.
- 6) Disable CDP and enable LLDP globally.

Phase 2: Layer 2 Configuration

- 1-2: Layer 2 EtherChannel
- 3-5: Configuring VLANs
- 6-8: VLAN Trunking
- 9-13: Configuring Access Ports and Special Ports
- 14-18: Layer 2 Security
- 19-20: Spanning Tree Protocol

EtherChannel

- 1) Configure L2 EtherChannel named **"Port-channel 1"** between the Vancouver Distribution switches. Use the **standard protocol**, and use **active negotiation** for both sides.
- 2) Configure L2 EtherChannel named **"Port-channel 1"** between the Burnaby Distribution switches. Use the **Cisco proprietary protocol**, and use **active negotiation** for both sides.

Configuring VLANs

- 3) Create and name the following VLANs on ALL **Distribution** and **Access** switches in **Vancouver**.

- **ASW10:** 100-104,99
 - **VLAN100:** VAN-FL10-PCs
 - **VLAN101:** VAN-FL10-VoIP
 - **VLAN102:** VAN-FL10-WiFi
 - **VLAN103:** VAN-FL10-Printers
 - **VLAN104:** VAN-Servers
 - **VLAN99:** VAN-Management
- **ASW11:** 110-113,99
 - **VLAN110:** VAN-FL11-PCs
 - **VLAN111:** VAN-FL11-VoIP
 - **VLAN112:** VAN-FL11-WiFi
 - **VLAN113:** VAN-FL11-Printers
 - **VLAN99:** VAN-Management

- **ASW12:** 120-123,99
 - **VLAN120:** VAN-FL12-PCs
 - **VLAN121:** VAN-FL12-VoIP
 - **VLAN122:** VAN-FL12-WiFi
 - **VLAN123:** VAN-FL12-Printers
 - **VLAN99:** VAN-Management
- Distribution switches should have **ALL VLANs** from all access switches.

4) Create the following VLANs on ALL Distribution and Access switches in **Burnaby**.

- **ASW8:** 80-83,99
 - **VLAN80:** BBY-FL8-PCs
 - **VLAN81:** BBY-FL8-VoIP
 - **VLAN82:** BBY-FL8-WiFi
 - **VLAN83:** BBY-FL8-Printers
 - **VLAN99:** BBY-Management
- **ASW9:** 90-93,99
 - **VLAN90:** BBY-FL9-PCs
 - **VLAN91:** BBY-FL9-VoIP
 - **VLAN92:** BBY-FL9-WiFi
 - **VLAN93:** BBY-FL9-Printers
 - **VLAN99:** BBY-Management
- Distribution switches should have **ALL VLANs** from all access switches.

5) Create the following VLANs on **Richmond**'s only access switch.

- **ASW1:** 10-13,99
 - **VLAN10:** RCH-PCs
 - **VLAN11:** RCH-VoIP
 - **VLAN12:** RCH-WiFi
 - **VLAN13:** RCH-Printers
 - **VLAN99:** RCH-Management

VLAN Trunking

6) Configure as **802.1Q TRUNK:** ALL Distribution-Access link interfaces, ALL L2 EtherChannel interfaces.

- Disable DTP on all trunk ports
- Use **Native VLAN 999**

7) Only allow VLANs on each trunk interface specific to the VLANs required by the access switch.

- **Vancouver DSW interface** <-> **VAN-ASW10:** 100-104,99,999
- **Vancouver DSW interface** <-> **VAN-ASW11:** 110-113,99,999
- **Vancouver DSW interface** <-> **VAN-ASW12:** 120-123,99,999
- **Burnaby DSW interface** <-> **BBY-ASW8:** 80-83,99,999
- **Burnaby DSW interface** <-> **BBY-ASW9:** 90-93,99,999
- **(VAN and BBY) DSW1** <-> **DSW2:** ALL VLANs

- 8) Configure the **Richmond** switch's router link as a TRUNK.
- Allow VLANs 10-13,99,999

Configuring Access Ports and Special Ports

- 9) Configure each used port from all access switches
- Manually configure access mode, Disable DTP
 - Match the VLAN with the corresponding endpoint. Apply the voice VLAN if needed to.
 - Ignore LWAP links for now (done in step 10 and 11)
- 10) **Vancouver** LWAPs will **NOT** use FlexConnect
- Configure the connections as an **access port**
 - Configure **VLAN 99** for this access port
- 11) **Burnaby** and **Richmond** LWAPs will use FlexConnect
- Configure the connections as a **trunk port**
 - Configure the **Wi-Fi and Management VLAN** for the trunk port. (*The Wi-Fi VLAN number will depend on the floor*)
 - **Native VLAN 99**
 - **Allowed VLANs:** Wi-Fi,99
- 12) Configure **VAN-DSW1's** connection to **WLC**
- Must support only **Wi-Fi** and **Management** VLANs (trunk)
 - Management VLAN should be **untagged**
 - Disable DTP
- 13) Configure all connections to the **Management PC**
- Access port, VLAN 99, DTP disabled

Layer 2 Security

- 14) Put all unused ports on a **Parking VLAN** on **ALL access and distribution switches** in the organisation, then disable them.
- First, create and name a Parking VLAN. Use VLAN 555.
- 15) Configure Port Security on each access port from **every access switch**.
- Make sure that the access port is explicitly in access mode.
 - Allow the **MINIMUM** necessary number of MAC addresses
 - PC links: 2
 - FlexConnect APs: 20
 - Everything else: 1
 - Use a violation mode that **BLOCKS and LOGS** invalid traffic
 - **Save learned MAC addresses** to running-config
- 16) Configure **DHCP Snooping** on ALL access switches
- Enable for **ALL ACTIVE VLANs**
 - Trust appropriate ports
 - Disable information of **DHCP Option 82 (relay agent information option)**
 - Rate limit of **10 pps** on untrusted ports

- 17) Configure **Dynamic ARP Inspection** on ALL access switches
- Enable for **ALL ACTIVE VLANs**
 - Trust appropriate ports
 - Enable ALL optional validation checks
- 18) **Disable LLDP transmit** on all access ports on all access switches.

Spanning Tree Protocol

- 19) Configure Rapid PVST+ on ALL access and distribution switches

VANCOUVER:

- **VAN-DSW1** is the primary root bridge for **Floor 10 and Management VLANs**
- **VAN-DSW2** is the primary root bridge for **Floor 11 and 12 VLANs**

BURNABY:

- **BBY-DSW1** is the primary root bridge for **Floor 8 and Management VLANs**
- **BBY-DSW2** is the primary root bridge for **Floor 9 VLANs**
- Use the **LOWEST** increment (0) for the primary root bridge
- Switches must be the **SECONDARY root bridge** for VLANs it does not cover on its primary root bridge

- 20) Enable **PortFast** and **BPDU Guard** on all ports connected to end hosts.
- Include the WLC connection as a **PortFast trunk**. Do NOT configure BPDU Guard on this port.
 - **HINT:** On access switches, there's a faster way to enable PortFast and BPDU Guard on all access ports. It only takes two commands.

Phase 3: Layer 3 Configuration (IPv4)

- 1: Layer 3 EtherChannel
- 2-9: IPv4 Addressing
- 10-11: Hot Standby Router Protocol
- 12: Access Control Lists

Layer 3 EtherChannel

1) Configure Layer 3 EtherChannel named "**Port-channel 1**" between the **Vancouver** Core Switches. Use the **standard** protocol, and use active negotiation for both sides.

IPv4 Addressing

2) Configure IP addresses in ALL ROUTERS

VAN-RTR:

- Loopback0: 50.50.1.1/32
- Loopback1: 50.50.10.10/32
- G0/0/0 (to VAN-CSW1): 192.168.50.1/30
- G0/0/1 (to VAN-CSW2): 192.168.50.5/30
- S0/1/0 (to BBY-RTR): 192.168.0.6/30
- S0/1/1 (to RCH-RTR): 192.168.0.9/30
- S0/2/0 (to ISP): 20.26.50.253/30

BBY-RTR:

- Loopback0: 51.51.1.1/32
- Loopback1: 51.51.10.10/32
- G0/0/0 (to BBY-DSW1): 192.168.51.1/30
- G0/0/1 (to BBY-DSW2): 192.168.51.5/30
- S0/1/0 (to VAN-RTR): 192.168.0.5/30
- S0/1/1 (to RCH-RTR): 192.168.0.1/30

RCH-RTR:

- Loopback0: 52.52.1.1/32
- Loopback1: 52.52.10.10/32
- G0/0/0: (sub-interfaces)
 - G0/0/0.10: 10.52.10.254/24
 - G0/0/0.11: 10.52.11.254/24
 - G0/0/0.12: 10.52.12.254/24
 - G0/0/0.13: 10.52.13.254/24
 - G0/0/0.99: 10.52.99.254/24
- S0/1/0 (to VAN-RTR): 192.168.0.10/30
- S0/1/1 (to BBY-RTR): 192.168.0.2/30

3) Enable IPv4 routing in ALL Core and Distribution switches (integrate this in your tasks 4 and 5 config)

4) Configure IP addresses in **Vancouver's Core and Dist Switches**. An **SVI interface** must be created in the Dist Switches for every production and management VLAN.

VAN-CSW1:

- Loopback0: 50.50.2.1/32
- Loopback1: 50.50.20.10/32
- Po1: 192.168.50.9/30
- G1/0/13 (to VAN-DSW1): 192.168.50.21/30
- G1/0/14 (to VAN-DSW2): 192.168.50.13/30
- G1/0/24 (to VAN-RTR): 192.168.50.2/30

VAN-CSW2:

- Loopback0: 50.50.2.2/32
- Loopback1: 50.50.20.20/32
- Po1: 192.168.50.10/30
- G1/0/13 (to VAN-DSW2): 192.168.50.25/30
- G1/0/14 (to VAN-DSW1): 192.168.50.17/30
- G1/0/24 (to VAN-RTR): 192.168.50.6/30

VAN-DSW1:

- Loopback0: 50.50.3.1/32
- Loopback1: 50.50.30.10/32
- G1/0/13 (to VAN-CSW1): 192.168.50.22/30
- G1/0/14 (to VAN-CSW2): 192.168.50.18/30
- SVI: 10.50.x.252/24 (x=VLAN ID)

VAN-DSW2:

- Loopback0: 50.50.3.2/32
- Loopback1: 50.50.30.20/32
- G1/0/13 (to VAN-CSW2): 192.168.50.26/30
- G1/0/14 (to VAN-CSW1): 192.168.50.14/30
- SVI: 10.50.x.253/24 (x=VLAN ID)

5) Configure IP addresses in **Burnaby's Dist Switches**. An **SVI interface** must be created in the Dist Switches for every production and management VLAN.

BBY-DSW1:

- Loopback0: 51.51.3.1/32
- Loopback1: 51.51.30.10/32
- G1/0/24 (to BBY-RTR): 192.168.51.2/30
- SVI: 10.51.x.252/24 (x=VLAN ID)

BBY-DSW2:

- Loopback0: 51.51.3.2/32
- Loopback1: 51.51.30.20/32

- **G1/0/24 (to BBY-RTR):** 192.168.51.6/30
- **SVI:** 10.51.x.253/24 (**x**=VLAN ID)

6) Configure IPv4 addresses in Vancouver's Servers:

- **VAN-SRV1:** 10.50.104.1/24
- **VAN-SRV2:** 10.50.104.2/24
- **Gateway:** 10.50.104.254

7) Configure a STATIC ADDRESS on the **WLC**

Configure the **Management** interface:

- **Address:** 10.50.99.10/24
- **Gateway:** 10.50.99.254
- **DNS Server:** 10.50.104.1 (**VAN-SRV1**)

8) Configure Management IP addresses on ALL access switches. Use the subnet's LAST usable address as gateway.

- **VAN-ASW10:** 10.50.99.1
- **VAN-ASW11:** 10.50.99.2
- **VAN-ASW12:** 10.50.99.3
- **BBY-ASW8:** 10.51.99.1
- **BBY-ASW9:** 10.51.99.2
- **RCH-ASW1:** 10.52.99.1

9) Park and disable ALL unused interfaces in **Vancouver's Core** switches

- Create a parking VLAN and assign the unused interfaces with the parking VLAN

Hot Standby Router Protocol

10) Configure HSRPv2 for **Vancouver's** subnets

- Match the active routers with their **primary root bridge VLANs** from Spanning Tree.
- Enable preemption
- Priority is **10 above default** for ACTIVE routers (110)
- Match the group numbers with the VLAN number
- Standby IP must be the **LAST USABLE ADDRESS**

(VAN-DSW1 primary root bridge: Floor 1 and Management)

(VAN-DSW2 primary root bridge: Floor 2 and 3)

11) Configure HSRPv2 for **Burnaby's** subnets

- Match the active routers with their **primary root bridge VLANs** from Spanning Tree.
- Enable preemption
- Priority is **10 above default** for ACTIVE routers (110)
- Match the group numbers with the VLAN number
- Standby IP must be the **LAST USABLE ADDRESS**

(BBY-DSW1 primary root bridge: Floor 1 and Management)

(BBY-DSW2 primary root bridge: Floor 2)

Access Control Lists

12) Configure the following **NAMED ACLs** using the order provided:

Don't forget to apply the ACLs according to best practise.

- **ALLOW-MANAGEMENT (standard)**
 - Permit ONLY LOCAL management VLANs to ALL network hardware (only permit the Vancouver MGMT VLAN in Vancouver, etc.)
 - applied **INBOUND** on ALL virtual lines
- **VOIP-IN (extended)**
 - **PERMIT:** DHCP requests
 - **PERMIT:** All VoIP VLANs
 - **PERMIT:** HSRP protocol via 224.0.0.102 (UDP port 1985)
 - **PERMIT:** CUCME server (VAN-RTR's Loopback1 address)
 - **PERMIT:** SCCP (VoIP protocol; TCP port 2000)
 - **PERMIT:** SIP (VoIP protocol; UDP 5060, TCP 5060, TCP 5061)
 - **PERMIT:** RTP streams (VoIP protocol; UDP ports 16384-32767)
 - applied **INBOUND** on ALL VoIP SVIs
- **PRINTER-IN (extended)**
 - **PERMIT:** DHCP requests, DNS, Syslog, ICMP reply
 - **PERMIT:** HSRP protocol to 224.0.0.102 (UDP port 1985)
 - **PERMIT:** All TCP protocols, **established connections only**
 - applied **INBOUND** on ALL Printer SVIs
- **PC-and-WIFI-IN (extended)**
 - **PERMIT:** DHCP requests, DNS
 - **PERMIT:** HSRP protocol via 224.0.0.102 (UDP port 1985)
 - **DENY:** Management and VoIP VLANs
 - **DENY:** 192.168.0.0/16 range (these are used by routers solely for backbone routing)
 - **DENY:** All router Loopback addresses
 - **Implicit permit** for Internet access
 - applied **INBOUND** on ALL Employee PC and Wi-Fi SVIs

Phase 4: Layer 3 Configuration (IPv6)

➤ 1-4: IPv6 Addressing

0) Enable **IPv6 UNICAST ROUTING** on ALL network devices
(integrate this in your config)

IPv6 Addressing

1) Configure IPv6 addresses in ALL ROUTERS:
(Don't forget to **enable IPv6** on all interfaces)

VAN-RTR:

- Loopback0: 2001::1/128
- Link-local: fe80::1
- G0/0/0 (to VAN-CSW1): 2001:192:168:50::2/127
- G0/0/1 (to VAN-CSW2): 2001:192:168:50::4/127
- S0/1/0 (to BBY-RTR): 2001:192:168::5/127
- S0/1/1 (to RCH-RTR): 2001:192:168::6/127
- S0/2/0 (to ISP): 2001:db8:2026:50::6/64

BBY-RTR:

- Loopback0: 2001::2/128
- Link-local: fe80::2
- G0/0/0 (to BBY-DSW1): 2001:192:168:51::2/127
- G0/0/1 (to BBY-DSW2): 2001:192:168:51::4/127
- S0/1/0 (to VAN-RTR): 2001:192:168::4/127
- S0/1/1 (to RCH-RTR): 2001:192:168::2/127

RCH-RTR: (don't forget to configure dot1Q encapsulation !)

- Loopback0: 2001::3/128
- Link-local: fe80::3
- G0/0/0: (sub-interfaces)
 - G0/0/0.10: 2001:10:52:10::ffff/64
 - G0/0/0.11: 2001:10:52:11::ffff/64
 - G0/0/0.12: 2001:10:52:12::ffff/64
 - G0/0/0.13: 2001:10:52:13::ffff/64
- S0/1/0 (to VAN-RTR): 2001:192:168::7/127
- S0/1/1 (to BBY-RTR): 2001:192:168::3/127

2) Configure IPv6 addresses in **Vancouver's Core and Dist Switches**. An SVI interface must be created in the Dist Switches for every production VLAN. Do NOT make an SVI interface for Management VLAN.

VAN-CSW1:

- **Loopback0:** 2001::50:1/128
- **Link-local:** fe80::50:1
- **Po1:** 2001:192:168:50::6/127
- **G1/0/13:** 2001:192:168:50::c/127
- **G1/0/14:** 2001:192:168:50::8/127
- **G1/0/24:** 2001:192:168:50::3/127

VAN-CSW2:

- **Loopback0:** 2001::50:2/128
- **Link-local:** fe80::50:2
- **Po1:** 2001:192:168:50::7/127
- **G1/0/13:** 2001:192:168:50::e/127
- **G1/0/14:** 2001:192:168:50::a/127
- **G1/0/24:** 2001:192:168:50::5/127

VAN-DSW1:

- **Loopback0:** 2001::50:3/128
- **Link-local:** fe80::50:3
- **G1/0/13:** 2001:192:168:50::d/127
- **G1/0/14:** 2001:192:168:50::b/127
- **SVI:** 2001:10:50:x::fffd/64 (**x**=VLAN ID)

VAN-DSW2:

- **Loopback0:** 2001::50:4/128
- **Link-local:** fe80::50:4
- **G1/0/13:** 2001:192:168:50::f/127
- **G1/0/14:** 2001:192:168:50::9/127
- **SVI:** 2001:10:50:x::fffe/64 (**x**=VLAN ID)

3) Configure IPv6 addresses in **Burnaby's Dist Switches**. An SVI interface must be created in the Dist Switches for every production VLAN. Do not make an SVI interface for Management VLAN.

BBY-DSW1:

- **G1/0/24:** 2001:192:168:51::3/127
- **SVI:** 2001:10:51:x::fffd/64 (**x**=VLAN ID)
- **Loopback0:** 2001::51:1/128
- **Link-local:** fe80::51:1

BBY-DSW2:

- **G1/0/24:** 2001:192:168:51::5/127
- **SVI:** 2001:10:51:x::fffe/64 (**x**=VLAN ID)
- **Loopback0:** 2001::51:2/128

- **Link-local:** fe80::51:2

4) Configure IPv6 addresses in Vancouver's Servers:

- **VAN-SRV1:** 2001:10:50:104::1
- **VAN-SRV2:** 2001:10:50:104::2
- **Gateway:** 2001:10:50:104::ffff

Phase 5: Routing

- **1-2:** OSPFv2 and OSPFv3
- **3:** OSPF Authentication
- **4:** Static Routing

OSPFv2 and OSPFv3

1) Configure OSPFv2 and OSPFv3 on ALL ROUTERS

- LAN-facing interfaces will be **AREA 50/51/52** depending on the city router
- Router-facing interfaces will be **AREA 0 backbone**
- Use Process **2** for OSPFv2, **3** for OSPFv3
- Router ID matches the **Loopback0 IP** in v2
- Router ID matches the **Loopback1 IP** in v3
- OSPF is **ENABLED** and **PASSIVE** on **loopbacks**
- Set the OSPF cost reference bandwidth to **10 Gbps** on all processes
- **Summarise the LAN IPv4 networks (10.x.0.0/16)** for each city router with its assigned /16 network address.
- Use **interface mode** to enable OSPFv2 (do **NOT** use the "network" command)
- All connection between OSPF neighbours must use **point-to-point**
- Do **NOT** configure OSPF on VAN-RTR's ISP-facing interface

2) Configure OSPFv2 and OSPFv3 on ALL L3 SWITCHES

- Use **Area 50 (Vancouver)**, **Area 51 (Burnaby)**
- Use Process **2** for v2, **3** for v3
- Router ID matches the **Loopback0 IP** in v2
- Router ID matches the **Loopback1 IP** in v3
- OSPF is **ENABLED** and **PASSIVE** on **LOOPBACKS and SVIs** (except Management VLAN)
- Set the OSPF cost reference bandwidth to **10 Gbps** on all processes
- Use the **"network"** command to enable OSPFv2. Only include **INTERFACE IP ADDRESSES** on the list (do **NOT** use network ranges)
- All connection between OSPF neighbours must use **point-to-point** (doesn't work on L3 EtherChannels, just leave it default)

OSPF Authentication

3) Authenticate OSPFv2 in all neighbour-connected interfaces.

- Do **NOT** authenticate passive interfaces.
- Use **per-area** authentication
- Use MD5 hashing, key 1
- Use **Sp33drun** as your OSPF key.

*HINT: If you don't know your neighbour-connected interfaces, what's the command to check your OSPF neighbours and the **interface** associated with it?*

Static Routing

4) Configure ONE static default route (both IPv4 and IPv6) for **VAN-RTR's** ISP connection. Use the ISP's address as the next hop.

- Advertise the default route in the OSPF domain.

Phase 6: Network Services

- 1-2: Domain Name System
- 3-4: DHCPv4
- 5-7: Network Address Translation
- 8-9: Network Time Protocol
- 10-11: SNMP and Syslog

DNS

1) Configure **A** and **CNAME** DNS entries on **VAN-SRV1**:

- **canada.ca**: 160.106.123.29
- **vancouver.ca**: 10.50.104.1
- **toronto.ca**: 10.60.10.100
- **montreal.ca**: 10.70.10.100
- **www.canada.ca**: canada.ca
- **www.vancouver.ca**: vancouver.ca
- **www.toronto.ca**: toronto.ca
- **www.montreal.ca**: montreal.ca

2) Configure all routers and switches to enable domain lookup and use **VAN-SRV1** as their DNS server.

DHCPv4

3) Configure DHCPv4 server pools for **each VLAN*** on ALL ROUTERS.

- Exclude the **first 20 and last 14** usable addresses on ALL pools.
- DHCPv4 service must be **decentralised** across the three city routers. They must provide addresses ONLY to their own LAN.
- *= Do **NOT** make a DHCP pool for **FL10-Servers** in **VAN-RTR**.

Pool: *VLAN Name*

Default gateway: 10.x.x.254

Domain name: speedrun.ccna

DNS server: 10.50.104.1 (**VAN-SRV1**)

WLC: 10.50.99.10 (Option 43 in all Management VLANs)

4) Configure D-switches to RELAY DHCP messages to their city router's **Loopback0** address.

Network Address Translation

5) Configure static NAT on **VAN-RTR** to enable Internet hosts to access VAN-SRV1 via the public address **20.26.50.121**

6) Configure NAT pools on **VAN-RTR**. Only **PCs** and **Wi-Fi clients** can use the Internet.

- Use **standard numbered ACLs** to define the appropriate inside local (private) address ranges.
 - **1:** 10.50.100.0/24 (VAN-FL10-PCs)
 - **2:** 10.50.110.0/24 (VAN-FL11-PCs)
 - **3:** 10.50.120.0/24 (VAN-FL12-PCs)
 - **4:** 10.50.102.0/24 (VAN-FL10-WiFi)
 - **5:** 10.50.112.0/24 (VAN-FL11-WiFi)
 - **6:** 10.50.122.0/24 (VAN-FL12-WiFi)
 - **11:** 10.51.80.0/24 (BBY-FL8-PCs)
 - **12:** 10.51.90.0/24 (BBY-FL9-PCs)
 - **13:** 10.51.82.0/24 (BBY-FL8-WiFi)
 - **14:** 10.51.92.0/24 (BBY-FL9-WiFi)
 - **21:** 10.52.10.0/24 (Richmond-PCs)
 - **22:** 10.52.12.0/24 (Richmond-WiFi)
- Define NAT pools of inside global (public) addresses for the following:
 - **VAN-FL10-PCs:** 20.26.50.0/27
 - **VAN-FL11-PCs:** 20.26.50.32/27
 - **VAN-FL12-PCs:** 20.26.50.64/27
 - **VAN-FL10-WiFi:** 20.26.50.96/29
 - **VAN-FL11-WiFi:** 20.26.50.104/29
 - **VAN-FL12-WiFi:** 20.26.50.112/29
 - **BBY-FL8-PCs:** 20.26.50.128/28
 - **BBY-FL9-PCs:** 20.26.50.144/28
 - **BBY-FL8-WiFi:** 20.26.50.160/30
 - **BBY-FL9-WiFi:** 20.26.50.164/30
 - **Richmond-PCs:** 20.26.50.193/32
 - **Richmond-WiFi:** 20.26.50.194/32

7) Apply your NAT pools on **VAN-RTR** to enable all PCs and Wi-Fi clients across the organization to access the Internet.

- Map the ACL with the appropriate NAT pool. Some pools will use **Port Address Translation**, while some will use **Dynamic NAT (one public address per private address)**.
 - **Dynamic NAT:** All **Vancouver** and **Burnaby** PC networks
 - **PAT:** All **Wi-Fi** networks, **Richmond** PC network

Network Time Protocol

NTP on Cisco routers takes notoriously long to synchronise. It may not even work well, especially on Packet Tracer. If it does not work, that's okay.

8) Configure NTP on **VAN-RTR**

- **Stratum 5** NTP server
- **NTP Server:** 216.239.35.4
- **Authentication key:** number 1, MD5, password is "Sp33drun"

9) All routers and switches should use **VAN-RTR's** Loopback1 interface as their NTP server

- Authenticate **VAN-RTR** using key number **1**, MD5 hashing, password is **"Sp33drun"**

SNMP and Syslog

10) Configure the SNMP community string **"Speedrunning"** on all routers and switches. The string should allow GET messages, but not SET messages (**read-only!**)

11) Configure **syslog** on all routers and switches

- send syslog messages to **VAN-SRV2**
- **all severity levels** should be logged
- enable logging to the buffer. Reserve 8192 bytes of memory for the buffer.

Phase 7: Securing Management Infrastructure

- 1: Basic Device Hardening
- 2-6: AAA RADIUS Authentication for Console Login
- 7-10: SSH Local Authentication for VTY Login

Basic Device Hardening

1) Configure the following on all network devices:

- Secure privileged EXEC mode with the password **"speedrunning"**. If possible, use **type 9 hashing**. Otherwise, leave it default to **type 5 hashing**.
- Create a username **"runner"** with encrypted password **"iamspeed"**. Grant the user **full privileges** in EXEC mode. If possible, use **type 9 hashing**. Otherwise, leave it default to **type 5 hashing**.
- Lock out the device for **2 minutes** upon **3 failed attempts** within **1 minute**.
- **Excluding Vancouver access switches**, configure the console line to require local login, 30-minute inactivity timeout, and synchronous logging.

AAA RADIUS Authentication for Console Login

Packet Tracer does not support using a specific source interface for outgoing RADIUS packets; we cannot configure ALL network devices to use RADIUS. For the sake of configuration, we will only configure Vancouver Access Switches to use RADIUS.

Also the commands used to configure a RADIUS client in Packet Tracer are deprecated in the real world.

2) Configure the AAA service on **VAN-SRV1**.

Network Configuration

- Use the **hostname** as the Client Name for the access switches.
- Use the **Management IP address** as the Client IP for the access switches.
- **Server type:** RADIUS
- **Secret Key:** radiuspa55

User Setup

- **Username:** consoler
- **Password:** speediam

3) Enable AAA and configure **VAN-SRV1** as the RADIUS server on **Vancouver's access switches**. Don't forget to put the secret key.

4) Configure a named list called **CONSOLE-LOGIN** to authenticate logins using the RADIUS server. If not available, then use the local database.

5) Configure AAA authentication for **console login** using the named AAA authentication method.

6) Configure a 30-minute inactivity timeout and synchronous logging on the console line.

Make sure to test the AAA by logging out of the switch and logging in via console. It should now prompt you for a username and password. Enter the AAA username and password.

SSH Local Authentication for VTY login

Configure in ALL network devices

7) Generate cryptographic SSH keys using the **largest modulus size**.

8) EXPLICITLY enable **SSH version 2**.

9) Configure ALL VTY lines to only allow SSH and authenticate against the **local database**.

10) Configure a 30-minute inactivity timeout and synchronous logging on the VTY lines.

Phase 8: GRE over IPsec VPN

- 1-2: IKE Phase 1: ISAKMP
- 3-5: IKE Phase 2: IPsec
- 6: Configuring Tunnel Interfaces
- 7: Configuring Static Routes to the Remote LANs

The **Toronto** and **Montreal** routers are already pre-configured with GRE over IPsec VPN from their side. All that's left is for **VAN-RTR's** configuration to fully establish a secure GRE over IPsec VPN tunnel to **Toronto** and **Montreal's** internal LAN networks.

WAN IPs:

TOR-RTR:

- **WAN:** 20.26.60.253
- **Tunnel0 to VAN:** 1.1.1.2/30

MTL-RTR:

- **WAN:** 20.26.70.253
- **Tunnel0 to VAN:** 3.3.3.2/30

IKE Phase 1: ISAKMP

1) Define **ISAKMP policy 10** based on the HAGLE practise:

- **Hash:** SHA (there's no SHA256 in PT)
- **Authentication:** pre-share
- **DH group:** 5 (PT doesn't support group 20)
- **Lifetime:** 86400 seconds (1 day)
- **Encryption:** AES-256

2) Configure a pre-shared authentication key (a password) and bind it to public WAN IPs of **Toronto** and **Montreal**.

```
crypto isakmp key onemoretime address 20.26.60.253
```

```
crypto isakmp key onemoretime address 20.26.70.253
```

IKE Phase 2: IPsec

3) Define the IPsec transform set.

```
crypto ipsec transform-set AES esp-aes 256 esp-sha-hmac
```

4) Configure Crypto ACLs. **Permit** the following:

- **TOR-TUNNEL (extended)**
 - **Protocol:** GRE
 - **Source:** 20.26.50.253
 - **Destination:** 20.26.60.253
- **MTL-TUNNEL (extended)**
 - **Protocol:** GRE
 - **Source:** 20.26.50.253
 - **Destination:** 20.26.70.253

5) Create a crypto map to bind the Phase 2 components together. Apply this crypto map directly to the outbound WAN interface.

```

crypto map GRE 10 ipsec-isakmp
  match address TOR-TUNNEL
  set peer 20.26.60.253
  set transform-set AES
!
crypto map GRE 20 ipsec-isakmp
  match address MTL-TUNNEL
  set peer 20.26.70.253
  set transform-set AES
exit
!
int s0/2/0
  crypto map GRE
!
exit
do wr

```

Configuring Tunnel Interfaces

6) Create two tunnel interfaces.

Tunnel0 to **TOR-RTR**

- **Address:** 1.1.1.1/30
- **Tunnel source:** s0/2/0
- **Tunnel destination:** 20.26.60.253
- **Tunnel mode:** GRE over IP

Tunnel1 to **MTL-RTR**

- **Address:** 3.3.3.1/30
- **Tunnel source:** s0/2/0
- **Tunnel destination:** 20.26.70.253
- **Tunnel mode:** GRE over IP

Configuring Static Routes to the Remote LANs

7) Configure two static routes to Toronto and Montreal's internal networks respectively. Use the appropriate **remote tunnel interface IP address** as the next hop for each route.

Toronto: 10.60.0.0/16

Montreal: 10.70.0.0/16

Phase 9: Wi-Fi

- 1-2: Initialisation
- 3: Configuring Dynamic WLAN Interfaces
- 4-6: Configuring WLANs and WPA2 Security

Initialisation

1) The WLC is currently not initialised. You will have to set up an admin username, password and initial settings. Use Vancouver's Management PC to access the WLC.

- **URL:** http://10.50.99.10
- **User:** admin
- **Password:** Sp33drun

a) Set up your Controller

- **System name:** WLC
- **Management IP address:** 10.50.99.10
- **Default Gateway:** 10.50.99.254
- **Management VLAN ID:** 0 (leave default)

b) Create your wireless network

This is arbitrary. Just make up a dummy network to proceed with the initialisation.

- **Network name:** dummy
- **Passphrase:** Sp33drun
- Ignore "3) Advanced Settings"
- Apply and wait for a minute

2) Browse "**https://10.50.99.10**" in your management PC's web browser, and log in with your credentials.

Configuring Dynamic WLAN Interfaces

3) Configure a dynamic WLAN interface for **all Vancouver Wi-Fi WLANs**

Controller > Interfaces > New

- **Interface Name:** *Wi-Fi VLAN name*
- **VLAN:** *Wi-Fi VLAN number*
- **Port:** 1
- **IP address:** 10.50.xx.10/24
- **Gateway:** 10.50.xx.254
- **DHCP Server:** 50.50.1.1

Configuring WLANs and WPA2 Security

Packet Tracer does not support WPA3 security. That's totally fine. You can still configure WPA2 security.

4) Configure a WLAN for **all Vancouver Wi-Fi WLANs**

WLANs > Go

- **Profile Name:** *Wi-Fi VLAN name*
- **SSID:** *Wi-Fi VLAN name*
- **ID:** *leave it as it is, it's arbitrary*

General Tab

- **Status:** Enabled
- **Interface:** *Match the appropriate interface*

Security Tab

- **Layer 2 Security:** WPA2 w/ AES, PSK: cisco123

5) Verify that all Vancouver LWAPs are associated with the WLC

Wireless > All APs

- If you do not see any online LWAPs, make sure they have an IP address via DHCP.

6) Test if you can access Wi-Fi from one of the Vancouver Laptops. Enter the appropriate WLAN credentials.

- You might connect to an AP that isn't from the same floor.
- PT has limitations with Wi-Fi DHCP, so you're going to get a management IP address. But that's okay, since the configuration is still correct.

Appendix: Troubleshooting commands for each phase

Use these commands to verify configurations and troubleshoot certain issues for each phase. Type a question mark "?" after the root show command to show more valid and specific commands.

Phase 1

```
show run | inc hostname|domain|lookup|password|banner
show cdp
show lldp
show lldp neighbors
```

Phase 2

```
show int status
show ip int bri

show etherchannel summary
show vlan bri
show int trunk

show port-security
show port-sec int <interface>
show port-sec address

show ip dhcp snooping
show ip dhcp snooping binding

show ip arp inspection
show ip arp insp statistics
show ip arp insp interfaces

show spanning-tree summary
show span vlan <vlan-number>
show span int <interface> detail
```

Phase 3

```
show ip int bri
show stand bri
show etherc sum
show access-lists
ping <ip-address>
```

Phase 4

```
show ipv6 int bri
ping <ipv6-address>
```


Phase 5

```
show ip route
show ip route static
show ip route ospf

show ip ospf
show ip ospf neighbor
show ip ospf int bri
show ip ospf int <interface>

show ipv6 route
show ipv6 route ospf
show ipv6 static

show ipv6 ospf
show ipv6 ospf nei
show ipv6 ospf int bri
show ipv6 ospf int <interface>
```

Phase 6

```
show run | inc domain|lookup

show ip dhcp pool
show ip dhcp binding

show ip nat statistics
show ip nat translations
show access-lists

show ntp associations
show ntp status

show run | sec logging
```

Phase 7

```
show run | i username|secret
show run | i login block-for
show run | s line con 0
show user
show login

show run | i aaa|radius

show run | s line vty
show ssh
show ip ssh
show crypto key mypubkey rsa
```

Phase 8

```
show crypto isakmp sa
show crypto isakmp policy
show crypto ipsec sa
show crypto map

show ip int b | int Tunnel
show int tunnel 0
show int tunnel 1
show ip route static
```